EA-MT Entity-Aware Machine Translation in English to Chinese

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Abstract

The primary objective of this project is to tackle the challenge of translating a given input sentence from the source language, English, into the target language, Chinese. This task focuses on addressing the accurate translation of named entities, which may be rare, ambiguous, or unknown to traditional machine translation systems.

The goal is to develop a machine translation system capable of effectively handling such named entities within input sentences, ensuring that their translation into the target language is precise.

1 Introduction

Named entity translation presents a unique set of challenges in machine translation systems due to their reliance on contextual, structural, and linguistic nuances. Ambiguity in context is a pervasive issue, as entities often depend on surrounding text for proper interpretation. For instance, the term "*Apple*" may refer to the fruit or the technology company, and distinguishing between these interpretations becomes particularly challenging when the contextual clues are limited or absent.

Another complexity arises from multi-word entities, such as names of organizations, products, or events, which require precise handling to maintain semantic accuracy. Literal translations of such entities often lead to awkward or incorrect results. For example, translating "*Department of Justice*" word-for-word fails to convey the appropriate equivalent term in the target language, thus diminishing the overall quality of the translation.

Cross-lingual consistency further complicates the translation process, as entities may have multiple valid representations in the target language. Maintaining consistency across translations is critical to ensuring that the intended meaning is preserved in various linguistic contexts.

To address these challenges, this study employs the **IWSLT 2017 English-to-Chinese** dataset, focusing on named entity translation. Named entities in the dataset are first identified using **SpaCy**, a robust named entity recognition (NER) tool. These recognized entities are then incorporated into a machine translation pipeline based on the **mBERT** architecture, leveraging pre-trained and fine-tuned weights to enhance the translation accuracy of complex and ambiguous entities. This methodology seeks to establish a robust framework for improving named entity translation, ensuring semantic and contextual fidelity in cross-lingual tasks.

2 IWSLT Dataset

The International Conference on Spoken Language Translation (IWSLT) 2017 dataset involves English-to-Chinese translations and provides a valuable resource for real-world conversational examples. For instance:

> *"Thank you so much, Chris. And it's truly a great honor to have the opportunity to come to this stage twice; I'm extremely grateful."*
>
> "非常谢谢，克里斯。的确非常荣幸能有第二次站在这个台上的机会，我真是非常感激。"

In this dataset, English sentence lengths are more dispersed compared to Chinese sentences. Chinese sentence lengths are concentrated

in shorter intervals because Chinese is a character-based language and typically uses fewer words.

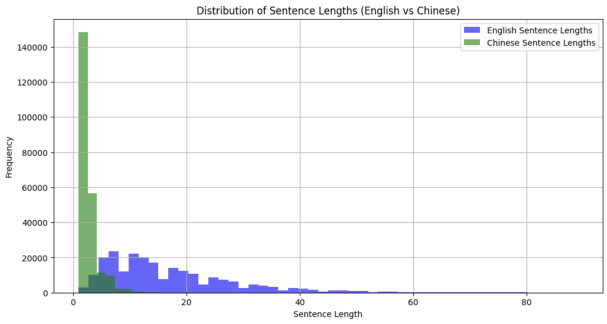


图 1: Distribution of Sentence Lengths (English vs. Chinese)

2.1 Linguistic Structural Differences

- **Character-Based vs. Word-Based Languages:**

  Chinese, as a character-based language, treats each character as the smallest semantic unit. In contrast, English, as a word-based language, requires additional auxiliary words (e.g., articles, prepositions, auxiliary verbs) to form complete sentences. Common examples of auxiliary words in English include articles (*"the"*), prepositions (*"in"*), and auxiliary verbs (*"is"*).

  For instance:

  – English: *"The dog is on the mat."* (6 words)

  – Chinese: "狗在墊子上。" (5 characters)

  This demonstrates that Chinese often uses fewer units to convey the same meaning, while English relies on additional grammatical components to structure sentences.

- **Differences in Word Order and Syntax:**

  English generally follows a fixed Subject-Verb-Object (SVO) structure, which makes sentence construction predictable and uniform. On the other hand, Chinese exhibits more flexibility in word order, allowing sentence structure to vary depending on the intended emphasis. However, this flexibility in Chinese may lead to alignment challenges during translation, especially when attempting to map flexible Chinese structures to the fixed SVO patterns of English.

  For example:

  – English: *"The red book on the table belongs to me."*

  – Chinese: "桌上的紅色書是我的。" (Reordered emphasis on *"the red book on the table"*)

  Furthermore, Chinese allows for reordering of elements to emphasize different parts of a sentence:

  – Standard: "我喜歡這本書。" (Subject-first: *"I like this book."*)

  – Emphasis: "這本書我喜歡。" (Emphasizing *"this book"*)

These structural differences pose challenges for machine translation models. While English requires adherence to strict grammatical structures, Chinese translation must account for its compact nature and flexible word order. Misalignment of sentence structures during translation can result in semantic or syntactic inaccuracies. Models must, therefore, be designed to accommodate the unique linguistic characteristics of both languages to achieve high-quality translations.

IWSLT English Word Cloud of Tokens in the Dataset

The word cloud illustrates the most common tokens within the IWSLT English-to-Chinese translation dataset. As expected, function words such as "the," "and," "to," "of," and "a" are prominently featured due to their frequent usage in the English language. This observation aligns with Zipf's Law, which states that in most natural language texts, function words tend to appear far more frequently than semantic words.

Furthermore, the word cloud highlights tokens such as "can," "they," and "this," which suggest the presence of conversational or descriptive text within the dataset. Although function words dominate the visualization, less frequent tokens like "what," "years," and "think" indicate potential narrative or explanatory content.

图 2: Word Cloud of Tokens in the IWSLT Dataset

IWSLT Chinese Word Cloud of Tokens in the Dataset

As shown in the visualization, high-frequency tokens such as “的,” “是,” “了,” “在,” and “有” dominate the dataset. These words are function words in Chinese and play a key role in sentence structure, similar to their English counterparts like “the,” “and,” or “to.” This observation aligns with Zipf’s Law, which posits that functional words frequently outnumber semantic words in natural language texts.

The visualization also highlights additional words like “我们,” “他们,” and “自己,” which indicate conversational or narrative contexts within the dataset. While function words are predominant, other tokens such as “因为,” “可以,” and “为什么” suggest the presence of explanatory or argumentative content.

图 3: Word Cloud of Tokens in the IWSLT Dataset (Chinese)

The analysis reveals that both character-based and word-based languages follow Zipf’s Law. This law highlights that in natural language, a small number of high-frequency tokens dominate the dataset, while the majority of tokens appear infrequently. The word cloud further reflects the diversity of the IWSLT dataset, suggesting that it likely contains a mix of narrative, conversational, and explanatory text types. The analysis reveals that both character-based and word-based languages follow Zipf’s Law. This law highlights that in natural language, a small number of high-frequency tokens dominate the dataset, while the majority of tokens appear infrequently. This visualization indicates that the dataset includes diverse text genres, such as narrative, conversational, and explanatory elements. It offers a preliminary understanding of the dataset’s composition and variety. It can guide preprocessing strategies, such as removing stopwords (e.g., “The", "a",“的,” “是,” “了,”) to focus on more semantically meaningful tokens for downstream natural language processing tasks.

In the Figure(4) we can see the complex sentence type appears most frequently. Complex sentences are typically sentences with one independent clause and one or more dependent clauses, often involving subordinating conjunctions like "because," "although," or "since.". Compound sentences are the second most frequent type it consist of two or more independent clauses connected by coordinating conjunctions such as "and," "but," or "or."

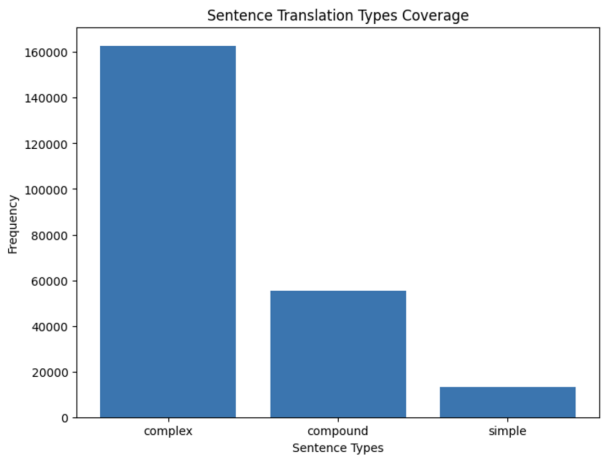
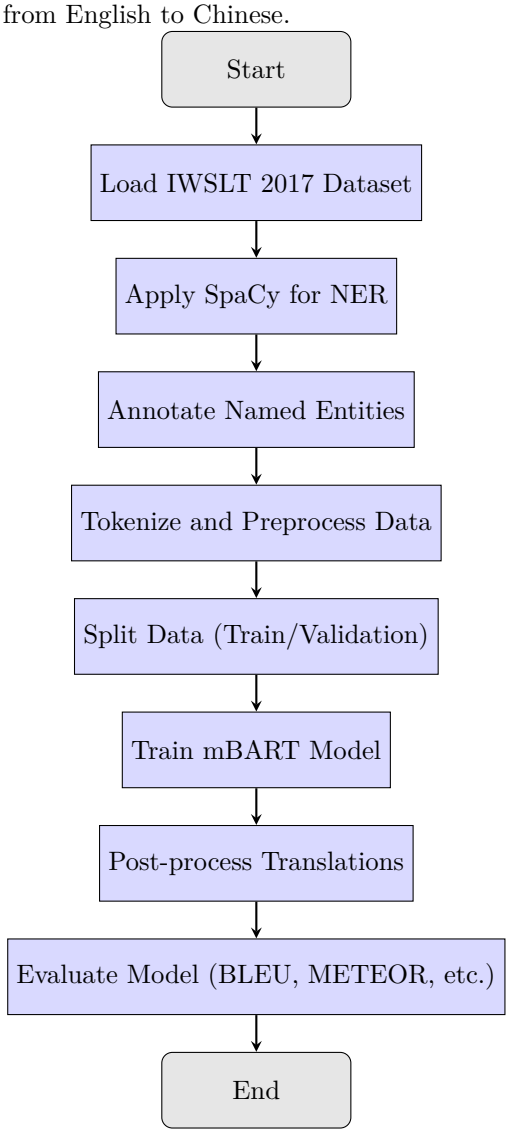


图 4: The Translation Coverage Types in IWSLT Dataset

The high frequency of complex sentences mean the translation model to handle subordinate clauses effectively.

3 Methodology - Flowchart

The process of integrating entity-aware translation is divided into several steps to manage the complexities of accurately translating named entities from English to Chinese.



In this project, I utilized SpaCy to extract named entities from English sentences and customized the entity annotation format (e.g., «`ENTITY:TEXT`»). This approach enhances the model's ability to recognize and handle named entities effectively. Next, the pre-annotated entity data was used as input for the mBART model. During training, the mBART model was fine-tuned with a focus on high-frequency entities such as `CARDINAL`, `DATE`, and `PERSON`. After translation, the entity annotations in the translated results were restored to readable text, ensuring that the output was both fluent and accurate. Multiple evaluation metrics, including METEOR and ROUGE, were employed to assess the model's performance in translating entities and maintaining contextual accuracy.

My strategy successfully combined the advantages of entity-aware annotations and a multilingual translation model, addressing the core challenge of named entity translation in English-to-Chinese tasks. This method significantly improved the model's understanding and translation accuracy of entities, achieving a strong translation score (METEOR 0.6649). In the following sections, I will provide a step-by-step explanation of each stage of the process.

4 Data Preprocessing

In this study, the **IWSLT** 2017 English-to-Chinese translation dataset was utilized, with a focus on addressing the challenge of translating named entities. To enhance the model's ability to process named entities, additional steps were taken to improve entity recognition. Specifically, **SpaCy's** Transformer-based English named entity recognition model (`en_core_web_trf`) was employed to identify entities within the source sentences. The identified entities were annotated in the format «`{ent.label}:{ent.text}>` , for example, «`ORG:Apple`». This annotation allows the model to recognize these key segments during training and inference. This preprocessing step significantly enhances the model's awareness of named entities, thereby optimizing translation quality.

5 Named Entity Recognition

Named entities can be conceptualized as specific instances of broader entity categories (for example, "Apple" is an instance of the category `Organization`). In the domain of information extraction, a named entity refers to a real-world object or concept that can be identified by a proper name, such as a person, location, organization, or product. These entities can exist either in physical form or in abstract form. Recognizing named entities is crucial for tasks such as machine translation, where the accurate identification and handling of named entities can significantly enhance translation precision.

5.1 SpaCy: `en_core_web_trf`

SpaCy provides pre-trained models for multiple languages. For this study, I selected the `en_core_web_trf` model to identify all named entities in the IWSLT dataset. This model leverages the Transformer architecture, which significantly improves the accuracy of named entity recognition, particularly in contexts with complex linguistic structures.

5.2 Named Entity Tags and Descriptions

The `en_core_web_trf` model supports the following entity tags:

Using SpaCy, I employed the `.label_` attribute to extract named entities and implemented a custom Python function, `mark_entities()`, to format the extracted entities into a structured representation of the form «`{ent.label}:{ent.text}`>. This explicit tagging approach enhances the model's ability to recognize and handle named entities during the translation process. By explicitly marking entities, the translation model gains a better understanding of their semantic significance, reducing the likelihood of misinterpretation or loss of meaning during translation.

5.3 Named Entity Tags Distributiion in IWSLT Dataset

In Figure(5), it can be observed that `CARDINAL` (numerical data) and `DATE` (dates) are the most frequently occurring entity types. This highlights the importance of numbers and dates in the dataset, such as references to years, statistics, or expressions of time. `PERSON` (names of people) and `GPE` (geopolitical entities like cities or countries) closely follow, emphasizing their critical role in improving Entity-Aware Machine Translation. These frequent entity types are vital for the model to learn and recognize entities effectively, which significantly improves the translation's accuracy.

Tag	Description	Example
PERSON	A person's name	"Barack Obama"
NORP	Nationalities, religious groups, or political groups	"American"
FAC	Buildings, landmarks, or facilities	"Eiffel Tower"
ORG	Organizations, companies, or institutions	"Google"
GPE	Countries, cities, or geopolitical regions	"France"
LOC	Non-GPE locations, such as mountain ranges or bodies of water	"Mount Everest"
PRODUCT	Objects, vehicles, or products	"iPhone"
EVENT	Named events	"Olympics"
WORK_OF_ART	Titles of books, songs, or artworks	"Mona Lisa"
LAW	Legal documents or clauses	"First Amendment"
LANGUAGE	Languages	"English"
DATE	Dates or time periods	"2024-01-01", "Wednesday"
TIME	Specific times	"2 PM", "noon"
PERCENT	Percentage expressions	"50%"
MONEY	Monetary amounts	"$20"
QUANTITY	Measurements	"10 kg"
ORDINAL	Ordinal numbers	"first", "2nd"
CARDINAL	Cardinal numbers	"one", "100"

表 1: Named entity tags, their descriptions, and examples in `en_core_web_trf`.

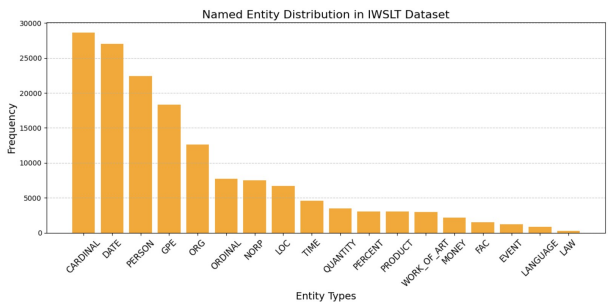


图 5: NER Bar Chart in the IWSLT Dataset

In Figure(6), the distribution of entity tags identified from the IWSLT dataset is shown. Most entities fall into ten primary categories. This further demonstrates that optimizing the model's Entity-Aware translation capabilities requires focused training on high-frequency types, such as `CARDINAL` and `DATE`.

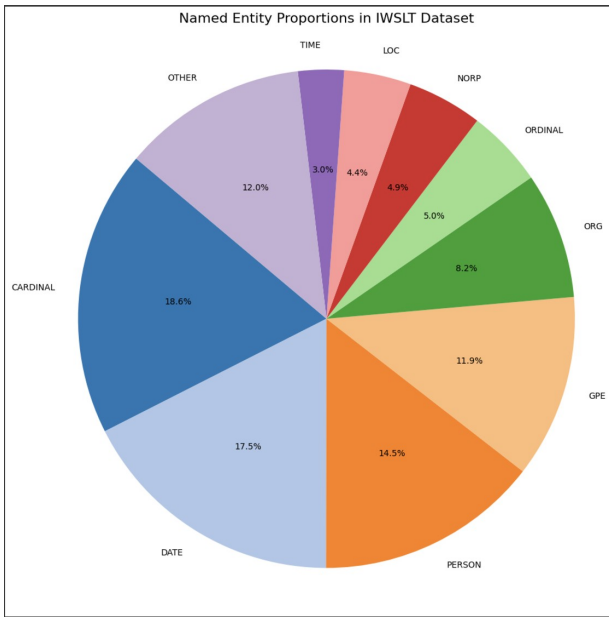


图 6: NER Pie Chart in the IWSLT Dataset

6 Model Design — mBART

The translation system was built upon the mBART model (`mbart-large-50-many-to-many-mmt`), a pre-trained Transformer-based sequence-to-sequence model designed for multilingual tasks. The tokenizer of the mBART model was customized to correctly recognize and process inputs with entity markers. By fine-tuning the preprocessed dataset, the model was able to leverage the additional contextual information provided by the named entity annotations, thereby improving translation performance.

未完成 🅕**明模型** 1. 詳細🅕明 mBART 的架構（Transformer-based），及其對多語言任務的適應性。2. 明預訓練權重的重要性及其在遷移學習中的優勢。3. 🅕**明參數**提供關鍵參數，例如學習率、批量大小、早停條件和訓練次數。解釋🅕什🅕使用交叉熵損失和標🅕平滑的原因，🅕🅕調其對模型穩定性的貢獻。4. 🅕**明權重**

5. **注意力分**🅕 **(Attention Visualization):** 用於分析模型如何處理語言結構，特🅕是命名實體。6. 🅕**明訓練過程**訓練環境: GPU 與訓練時間 7. 🅕**明 Loss**

8. **Precision Recall F1** 9. **模型比較** 1. 🅕有訓練過的 mBART 與訓練過的比較改進幅度，用不同的評估指標。2. mBART 與其他翻譯模型（如 Google NMT、Transformer-based 模型）的優劣勢。

7 Evaluation Metrics

We chose METEOR and BLEU for our evaluation metric, which assesses machine translation quality through unigram matching with human references. METEOR compares unigrams by surface form, stem, and meaning, calculating scores based on precision, recall, and fragmentation. And BLUE is One of the first metrics to correlate highly with human judgment, BLEU remains popular and cost-effective. However, BLEU does not account for intelligibility or grammatical correctness. Therefore, we use multiple evaluation methods to more truly reflect the translation capabilities of our model.

7.1 METEOR Calculation

METEOR (Metric for Evaluation of Translation with Explicit ORdering) is an automatic metric used for evaluating the quality of machine translation. Unlike BLEU, which focuses solely on precision, METEOR takes both precision and recall into account, providing a more balanced assessment of translation quality. The key steps involved in calculating METEOR are as follows:

1. **Tokenization and Alignment:**
   - Tokenize the hypothesis (translated text) and reference (ground truth) sentences into words.
   - Align the tokens in the hypothesis with the tokens in the reference based on exact matches, stemmed matches, synonyms, or paraphrases (if applicable).
2. **Calculation of Precision and Recall:**
   - Precision (`P`): The fraction of tokens in the hypothesis that are aligned with to-

kens in the reference.

$$P = \frac{\text{Number of Matched Tokens}}{\text{Total Tokens in Hypothesis}}$$

- Recall (`R`): The fraction of tokens in the reference that are aligned with tokens in the hypothesis.

$$R = \frac{\text{Number of Matched Tokens}}{\text{Total Tokens in Reference}}$$

3. **Harmonic Mean (F-score):** Combine precision and recall using a harmonic mean, which assigns greater weight to recall.

$$F_{\text{mean}} = \frac{(1+\beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

where $\beta = 3$, giving higher importance to recall.

4. **Fragmentation Penalty:** To penalize translations with disordered word sequences, a fragmentation penalty (`Pen`) is applied based on the number of contiguous segments in the alignment:

$$\text{Pen} = \gamma \cdot \left(\frac{\text{Chunks}}{\text{Matches}}\right)$$

where γ is a predefined constant, typically set to 0.5.

5. **Final METEOR Score:** The final METEOR score is computed as:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Pen})$$

This metric is particularly well-suited for evaluating translations that emphasize semantic similarity and word order. By incorporating both precision and recall, along with penalties for disordered sequences, METEOR provides a robust evaluation of translation quality.

7.1.1 Semantic Similarity in METEOR for Non-English Languages

METEOR 中的語意相似的功能只能使用在英文當中, 所以我使用了哈工大的資料集建立了同義詞的詞林，Ⓕ也使用了 OpenHowNet 來計算語意的相似度

7.2 Semantic Similarity in METEOR for Non-English Languages

METEOR is designed to support semantic similarity calculations through the inclusion of stemming, synonyms, and paraphrasing. However, these features are primarily optimized for English and may not perform as effectively for other languages, such as Chinese. To address this limitation, I incorporated additional resources to enhance semantic similarity detection for Chinese translations:

- **Synonym Forest from HIT:** I utilized a synonym lexicon built from the Harbin Institute of Technology (HIT) dataset. This lexicon provides a comprehensive collection of Chinese synonyms, allowing the METEOR metric to match semantically equivalent words in translations. For example, synonyms such as 好 (good) and 优良 (excellent) can be aligned during evaluation.

- **Semantic Similarity with OpenHowNet:** To further improve semantic similarity detection, I integrated `OpenHowNet`, a large-scale knowledge graph specifically tailored for semantic understanding in Chinese. OpenHowNet provides detailed semantic representations for Chinese words, which were used to compute similarity scores between non-exact matches. This approach ensures that even if two words are not exact synonyms, their semantic similarity can still be quantified and considered in the evaluation.

- **Implementation in METEOR:** The synonym and semantic similarity components were incorporated into the alignment phase of METEOR. During the alignment process:
  - Exact matches between tokens were prioritized.
  - If no exact match was found, tokens were checked for synonym matches using the HIT lexicon.
  - If neither an exact nor a synonym match was found, OpenHowNet was used to calculate a semantic similarity score. Tokens

with a similarity score above a predefined threshold (e.g., 0.8) were aligned.

This integration significantly enhanced the capability of METEOR to evaluate translations involving Chinese by capturing semantic equivalence more effectively. The combination of the HIT synonym lexicon and OpenHowNet ensured a robust semantic similarity mechanism, extending the functionality of METEOR beyond its native support for English.

7.3 BLUE